

Programming and Data Structures Laboratory (CS19003)

Assignment 1: Section 6

Department of CSE, IIT Kharagpur

14th January, 2026

General instructions to be followed strictly:

- Name your file as `<roll_no>_<assignment_no>_<question_no>`. For example, if your roll number is 14CS10001 and you are submitting assignment 3 question number (a), then name your file as 14CS10001_3_a.c
 - Write your name, roll number, and assignment number at the beginning of your program.
 - Use proper indentation in your code and comment. Marks may be deducted without proper indentation and comments.
 - Submitting the **correct file** is your responsibility, any request of re-submission after the lab will **never** be entertained.
 - **Zero tolerance** policy will be strictly followed for **plagiarism**. The marks will be put straight zero if such cases are identified.
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Question-1 (10 points)

Write a C program to determine the total amount accrued in a cumulative fixed deposit account for 3 years at the rate of 12 percent per annum if the interest is compounded annually on an initial principal amount of Rs. 1,00,000. The program should print out the total amount.

Question-2 (20 points)

In this problem, you need to evaluate trigonometric functions like $\sin(x)$ and $\cos(x)$, using their respective series expansion formulae (up to **first 4 terms**).

$$\sin(x) = x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \frac{1}{5040}x^7 + \dots$$

$$\cos(x) = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6 + \dots$$

Input an angle x (in radians) from the user.

(a) Compute approximate values of $\sin(x)$ and $\cos(x)$ using the series expansion formulae.

(b) Compute $\sin(2x) = 2\sin(x)\cos(x)$, using the values computed in part (a).

(c) Compute $\sin(2x)$ directly using the first four terms of the series expansion formula for $\sin(x)$ and compare the value with that obtained in (b).

Example:

The value of x (in radian): 0.5236
The value of $\sin(x)$ is 0.500001
The value of $\cos(x)$ is 0.866025
The value of $\sin(2x)$ using formula $2\sin(x)\cos(x)$ is 0.866026
The value of $\sin(2x)$ using series is 0.866022

Question-3 (20 points)

Declare four floating-point numbers, namely x_1, y_1, x_2, y_2 . Assume that they are the (x, y) coordinates of the endpoints of a straight line segment AB . Declare two more floating-point numbers, m and n . Take the values of these six numbers as input during execution. Compute the coordinates of the point P on the line segment AB , which internally divides AB in the ratio $m : n$. Print the coordinates of P .

Example:

Enter coordinates of A (x_1 y_1): 2 4
Enter coordinates of B (x_2 y_2): 8 10
Enter the ratio m and n (m n): 1 2
The coordinates of point P are: (4.000000, 6.000000)